Optimal Market Making in Intraday Electricity Markets via Hawkes Process Modeling

by

Robert Hennings

Submitted to the

Chair of Financial Economics at the

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Masterthesis

Master of Science in Quantitative Finance

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Abstract

DEEP Reinforcement (RL) is one of the most successful and recent techniques in the field of Artificial Intelligence (AI). It has been successfully applied to various fields, including robotics, gaming, and finance. In this paper, we explore the application of Deep RL in the context of algorithmic trading in the electricity market. We propose a novel approach that combines Deep RL with advanced neural network architectures to optimize trading strategies. Our method leverages historical market data and real-time information to make informed trading decisions. We evaluate our approach using a comprehensive dataset from the electricity market and compare it with traditional trading strategies. The results demonstrate that our Deep RL-based method outperforms conventional approaches, achieving higher returns and lower risk. This research contributes to the growing body of literature on AI applications in finance and provides valuable insights for practitioners in the energy trading sector.

Keywords: **Reinforcement Learning**, **Deep Learning**, **Electricity Market**, **Trading**, **Algorithmic Trading**, **Energy Trading**, **Financial Economics**, **Quantitative Finance**, **Machine Learning**, **Artificial Intelligence**, **Neural Networks**, **Time Series Analysis**, **Forecasting**, **Decision Making**, **Optimization**, **Stochastic Optimal Control**, **Market Making**, **Market Microstructure**,

JEL Classification: **C61**, **C63**, **G11**, **G17**, **Q41**

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Contents

List of Abbreviations

LOB	Limit Order Book
LO	Limit Order
MO	Market Order
OI	Open Interest
API	Application Programming Interface
MM	Market Maker
NHPP	Non-Homogeneous Poisson Process
ACF	Autocorrelation Function
PACF	Partial Autocorrelation Function
PDF	Probability Density Function
CDF	Cumulative Probability Density Function
ECDF	Empirical Cumulative Probability Density Function
QQ	Quantile-Quantile
MLE	Maximum Likelihood Estimation
RES	Renewable Energy Sources
EPEX SPOT	European Power Exchange Spot Market
EEX	European Energy Exchange
CID	Continuous Intraday
IDA	Intraday Auction
EEG	Erneuerbare Energien Gesetz
AI	Artificial Intelligence
RL	Reinforcement Learning
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
DDPG	Deep Deterministic Policy Gradient
PPO	Proximal Policy Optimization
A2C	Advantage Actor-Critic
A3C	Asynchronous Advantage Actor-Critic
TD3	Twin Delayed Deep Deterministic Policy Gradient
SAC	Soft Actor-Critic
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
CNN	Convolutional Neural Network

List of Symbols

t	discrete time index t
T	discrete terminal time index T
δ	order book depth
Δp	tick size
$P(\delta)$	fill-probability of order book depth
V	order volume
q	inventory level
$\mathcal{A}$	Set of Admissible Strategies
ϕ	(Running) Inventory penalty
λ	(Conditional) Intensity
$\mathcal{F}$	Filtration
S	Midprice
X	Cash-Process
W	Brownian Motion

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1 Introduction

The ever increasing power generation attributed to renewable energy sources (RES), predominantly in Germany within Europe, with a current share of 57,2 % of produced gross-electricity by RES (2025)[1], highlights the rapid decarbonization of the energy sector. By regulatory design, RES have a feed-in priority[2], meaning that they are guaranteed to be integrated into the electricity grid and are given priority over other sources of electricity generation, thereby directly impacting the price formation process.

As the RES production is heavily weather dependent, this uncertainty is transferred into the (european) electricity markets, resulting in increased price volatility and a growing demand for balancing the short term imbalances as well as forecasting errors in the continuous intraday markets (CID), where the financial trading of these positions can be carried out.

1.1 The financialization of the electricity market

In Europe, the main exchanges for trading electricity are the Intercontinental Exchange (ICE), the NordPool, the European Energy Exchange (EEX) and the European Power Exchange (EPEX SPOT). The latter one, acting as the reference spot market, where the short term trading of electricity takes place, what this thesis focuses on.

These exchanges provide a platform for trading electricity contracts, including futures, options, and spot contracts. The spot trading is organized in a CID market and three intraday auctions (IDA), where the CID allows for trading of electricity contracts with delivery within the next 24 hours, while the IDA is a single auction that takes place at a specific times each day, allowing for trading of electricity contracts with delivery on the following day. The CID market offers 1-hour contracts, 30-Minute contracts, and the most granular 15-Minute contracts, covering the entire 24-hour period of the next following day. The IDA, on the other hand, focuses on 1-hour contracts for the next day, providing a platform for market participants to adjust their positions based on updated information and forecasts. More details about the market structure and trading mechanics of the EPEX SPOT will be provided in chapter 2.1 The structure of the European Power Market.

The trading in these contracts is facilitated through a limit order book (LOB), where buyers and sellers submit their orders, which are then matched based on price and time priority. The LOB provides a transparent view of the market, allowing participants to see the available liquidity and make informed trading decisions. This *order-driven market* structure is a key feature of the electricity markets, as it allows for efficient price discovery and facilitates the trading of electricity contracts in a competitive environment, in

[1] See Data from [Arb26], last accessed: 2026-05-13 19:08.
[2] See [Eur18] for the relevant EU directive and [Eur24] for the support scheme recommendation, as well as the national EEG [Deu23].

contrast to a *quote-driven market* structure, where market makers provide liquidity by quoting buy and sell prices for the contracts. The dynamics of the LOB are influenced by various factors, including the arrival of limit orders and market orders, cancellations, and modifications of existing orders. In order for a smooth operating and liquid market, it is crucial to have a sufficient number of market participants, including so called market makers, who provide liquidity by continuously quoting buy and sell prices for the contracts, making sure incoming orders can be executed efficiently at all times. Liquidity provision is carried out by these market makers at a rapid speed pocketing the bid-ask spread, which is the difference between the highest price a buyer is willing to pay and the lowest price a seller is willing to accept. As the bid-ask spread is usually only a few cents wide, at best, earning this gap has to be done several thousands of times a day, which is only possible through the use of algorithmic trading strategies, which are often based on statistical and machine learning techniques, designed to quickly analyze huge amounts of market data, identify trading opportunities, and execute trades at high speeds. With an average daily order volume of 1,242,762 orders in the EPEX SPOT CID market in 2019, this leaves no real other options, as to make use of algorithmic trading strategies, to be able to compete in this market.[3] This process is essential for maintaining an efficient and liquid market, as it allows for quick execution of trades and helps to reduce price volatility. The presence of market makers also encourages more participants to enter the market, as they can be assured of liquidity and competitive pricing. In recent years, the role of market makers has become increasingly important in the electricity markets, with many participants often using algorithmic trading strategies, what is also evident in the share of submitted orders to the EPEX SPOT via Application Programming Interfaces (APIs), that is around 65% and represents around 47% of traded volume.[4]

The discrepancy between automatically and human interacted manual trading volume is due that the automatically submitted orders are often smaller in size, but more frequent, while the manually submitted orders are often larger in size, but less frequent, what later motivates modelling decisisons with respect to variable order sizes.

This financialization highlights the economic importance of efficient trading strategies and risk management techniques in the electricity markets, as market participants seek to optimize their returns while managing the risks associated with price volatility and uncertainty. The unique characteristics of the electricity markets, such as the non-storability of electricity and the influence of weather conditions on supply and demand, further emphasize the need for sophisticated modeling approaches that can capture these dynamics effectively. In this context, the application of Hawkes processes for optimal market making in intraday electricity markets offers a promising avenue for developing advanced trading strategies that can adapt to the complex and dynamic nature of these markets, what this thesis essentially covers.

Here the main overall approach resembles around the idea that the LOB can be treated

[3] 2019 figures, see [EPE19] for more details.

[4] 2019 figures, see [EPE19] for more details, as of 20-05-2026 455 Exchange Members are registered and 4XX APIs.

as stochastic system, also referred to as *zero-intelligence model* but will be enhanced by including some priors like non-constant order intensities or dependency structures [5], in contrast to the often used second main approach, treating the LOB as a strategic game between market participants, where the actions of one participant can influence the actions of others, leading to a complex interplay of strategies and counter-strategies, also known as *perfect rational agent-based model*. By modeling the LOB as a stochastic queueing system, we can also allow for flexible modelling techniques for the involved order types, that can be submitted **market orders** (MO), **Limit Orders** (LO) and **cancellations**, which are the main drivers of the LOB dynamics and are explained in more detail in 2.2.1 Order Types.[6]

Increasing trading volumes, Increasing renewables energy production, increasing uncertainty through renewables through weather induced volatility flexible planning and balancing through intraday market mechanism, cite volume increase, connection to financial markets and LOB, efficient price discovery, high share of algorithmic trading unique stylized facts that need to be modelled,

1.2 Main Research Objectives of the Thesis

The main objective of this thesis is to explore the application of Hawkes processes for optimal market making in intraday electricity markets. Specifically, we aim to develop a modeling framework that captures the dynamics of the limit order book (LOB) and the interactions between different order types, such as limit orders, market orders, and cancellations. By leveraging the self-exciting nature of Hawkes processes, we seek to model the arrival of orders and their impact on the LOB, allowing us to develop trading strategies that can adapt to the evolving market conditions. Through empirical analysis and numerical simulations, we aim to evaluate the performance of our proposed approach and compare it with traditional market making strategies, ultimately contributing to the advancement of trading techniques in the context of intraday electricity markets. Taking into account the often not properly considered dynamics of Cross excitation, time varying baseline intensities and general intensities, external excitement through updated forecasts and external variables, we seek to develop a comprehensive modeling framework that can effectively capture the complex dynamics of the LOB and provide insights into optimal market making strategies in the context of intraday electricity markets. The main research questions that this thesis seeks to address are:

- How can Hawkes processes be effectively applied to model the dynamics of the limit order book in intraday electricity markets?
- What are the key stylized facts of the limit order book in the EPEX SPOT CID market, and how can they be incorporated into the modeling framework?

[5] See [Jai+24].
[6] See [Gou+13].

- How does the performance of the proposed Hawkes process-based market making strategy compare to traditional market making strategies in terms of profitability and risk management?
- What are the implications of the findings for practitioners in the energy trading sector, and how can they be used to inform trading strategies and risk management techniques in intraday electricity markets?

In answering the above objectives, the main contributions are

This thesis is aimed at obtaining the optimal, dynamic quote distances from the best bid/ask using stochastic optimal control theory solving the Hamilton-Jacobi-Bellman (HJB) equation. This is done, proposing different, extending modelling frameworks, all making use of the multivariate marked Hawkes model with time-varying baseline intensities. The bid and ask side will be modelled separately, allowing for interactions, as well as per side, then the three main components, affecting price building, namely Market Order Arrivals, Limit Order Arrivals and Cancellations.By starting from this baseline model framework with six (interaction) channels, it will be extended to the main model by also incorporating the first three price levels per side. Two additional extensions will be reviewed, namely the *state-dependent* and the *marked* one. In the state-dependent extension, the intensities will be dependent on parameters like imbalance, spread, queue size, inventory, time to delivery and time of day. In the marked extension, order size and price impact will be used as marks, especially for MO and LO. A possible optional extension could be the use of Deep Reinforcement Learning (DRL), since especially in the state-dependent extensions, we face a rich state space, which can be hard to solve through traditional methods, but is well suited for DRL, which can be used to learn optimal policies. For the estimation strategy, the maximum likelihood estimation (MLE) will be the main workhorse, but we will consider also non-parametric spline based estimation for the excitation kernel functions.

1.3 Structure of the Thesis

The thesis is organised as follows, after having laid out the motivation for the chosen research topic along with its main research objectives and their added new insights in 1 Introduction the theoretical background follows. In chapter 2 Theoretical Background we provide the necessary theoretical details on the structure of the european power market, its organization, functional mechanics and the key problem of market making, what this thesis focuses on. Following in 3 Empirical LOB Characteristics we present the empirical characteristics, or *stylized facts*, of the LOB in the EPEX SPOT CID market, which act as the main orientation basis and benchmark for the modelling approach in the later chapters.[7] Being equipped with the needed background knowledge about the market mechanics and

[7] See [Jai+24].

typical patterns, the central modelling aspcets are covered in 4 Mathematical Modelling. There the Hawkes process, along with its underlying mathematical theory, is introduced and the modelling approach for optimal market making is discussed. As the proposed modelling is put to test in chapter 5 Numerical Analysis and Simulation being used on actual, empirical data from the EPEX SPOT CID market, where the performance of the optimal market making framework is evaluated, the discussion is carried out in chapter 6 Discussion. Final conclusions are drawn in chapter 7 Conclusion, where the main findings of the thesis are summarized, their implications are discussed, and avenues for future research are suggested.

2 Theoretical Background

We begin by introducing the structure of the European Power Market, with a particular focus on the EPEX SPOT market, which serves as the primary context for our study. We then delve into the organization of trading which is based on the LOB system, where the main order-variants and the matching mechanism will be discussed. The chapter will be closed by analysing the key problem of market making, which is the main focus of this work.

2.1 The structure of the European Power Market

The core structure of the european power market is fourfold, consisting of the Derivatives Market, the Day-Ahead or Forward Market, the Intraday Market, and the Balancing Market. Their (chronological) order can be seen in Figure 2.1, where a special emphasis is put on the timely ordering of the different markets and their events, as this plays a crucial role for modelling. Their order follows the time-to-delivery of the traded contracts, with the Derivatives Market being the furthest away from delivery, followed by the Day-Ahead Market, the Intraday Market, and finally the Balancing Market, which is closest to delivery. In the following, these components will be briefly discussed in their chronological

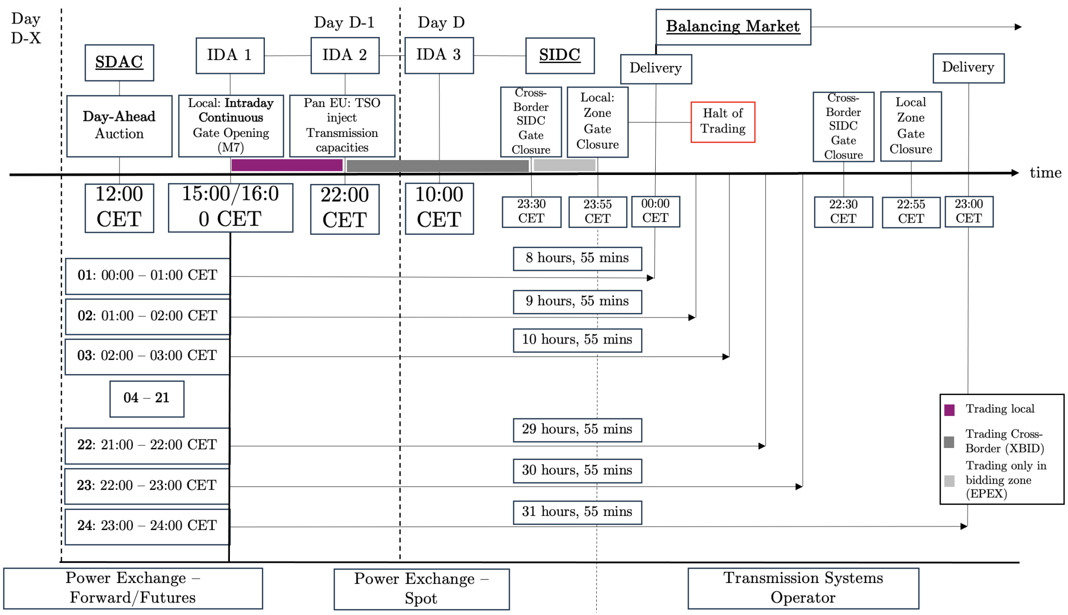
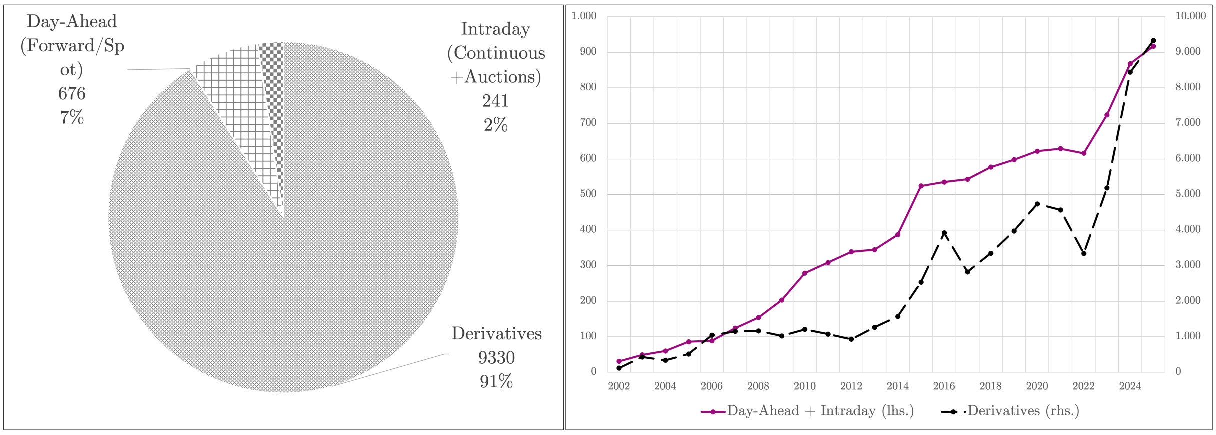


Figure 2.1: Schematic Overview of the European Power Market Design with exemplary hourly products in chronological order.[8]

order, that also resembles the markets importance based on the Trading Volume, that can be seen in Figure 2.2. In the left panel of Figure 2.2, we can see the annual Trading

[8] Based on [Kie19], as of 16.05.2026.
[9] Based on [Kie19], as of 16.05.2026.



Panel a): Annual Trading Volume in GWh of the year 2025, split into the core structural sub-markets. Panel b): Historic development of the trading volume for the two categories Derivatives Market and Spot Market, where the latter one includes the Day-Ahead and Intraday Market volumes.

Figure 2.2: European Power Market Trading Volume: Historic (panel b, right) and Year 2025 (panel a, left) Figures.[9]

Volume in GWh of the year 2025, split into the core structural sub-markets, where the clear dominance of the Derivatives Market is evident, where around 91% of traded volumes is concentrated. In the right panel of Figure 2.2, we can see the historic development of the trading volume for the two categories Derivatives Market and Spot Market, where the latter one includes the Day-Ahead and Intraday Market volumes. From the historic development, the nearly 10:1 relation of the volumes is verified.

2.1.1 The Derivatives Market

The derivatives market of the EEX is the most one in terms of trading volume, where the traded contracts are primarily settled financially, meaning that the contracts are not necessarily linked to a physical delivery of electricity, but rather serve as financial instruments for hedging and speculation over the medium term.[10] The product range includes futures, options, and other derivatives based on electricity prices, that can be traded for various delivery periods, such as respective next 11 days and the next 2 weekends, the current and the next 4 weeks, the current and the next 24 months, 11 quarters and 10 years. Exchange trading can be conducted from 8:00 am to 6:00 pm (CET).

2.1.2 The Day-Ahead Market

In contrast to the Derivatives Market of the EEX, where the traded contracts are primarily settled financially, a physical delivery is mandatory on the Day-Ahead Market (also referred to as *Spot Market*). The main purpose of the Day-Ahead Market is to already

[10] A full list of all tradable instruments and their specifications can be found here.

lock in certain volumes for the next day (D). Based on newly arrived forecasts, updated information and the need to balance out the supply and demand, market participants can adjust their positions in the Intraday Market. Market participants can submit so-called bidding curves, stating precisely how much power they would want to generate or consume at each price tick between the minimum and maximum prices at each operational subperiod of the next following day, i.e. 24 hours, 30 Minutes and 15 Minutes, resulting in 96 subperiods in total. Submitted prices at which the produces would want to sell power is by regulatory design set to the marginal cost of production, that dictates a natural merit-order hierarchy of the different production technologies. RES naturally operate at the lowest marginal costs, often assumed to be zero, therefore occupying the lowest price levels and the left side of the supply curve, while conventional power plants, such as coal, gas and nuclear, operate at higher marginal costs, occupying the higher price levels and the right side of the supply curve. These biddingcurves have to be submitted up until the gate closure time, which is typically around 12:00 CET on the day before delivery (D-1), allowing them to react to the latest information. Most general Market areas are further divided into bidding zones, which are defined by the transmission system operators (TSOs) and represent areas with similar supply and demand conditions. The geographical area of Germany is divided into four control areas, according to the four TSOs, which are TenneT (DE1), 50Hertz (DE2), Amprion (DE3) and TransnetBW (DE4), that are encoded in the send orders as can be seen in the data. The price formation is then carried out through a uniform price auction, where the market clearing price is determined by the intersection of the aggregated supply and demand curves, which are formed by the submitted orders (for every granularity of the next following day, i.e. 96 15-Min subperiods). The market clearing price is the price at which the total quantity of electricity demanded equals the total quantity supplied, and all transactions are executed at this price. It also serves as a reference price for the Intraday Market and the Balancing Market, as it reflects the market's expectations for the next day. All power producers that have submitted orders at or below the market clearing price will be scheduled to produce electricity, and are awarded the market clearing price for their production. An exemplary supply and demand curve for the market area DE-LU can be seen in Figure 2.3, for the delivery date of the 28th of May, 00:00-00:15. To enhance european coupling, optimizing cross border flow and ultimately welfare, the Day-Ahead Market is coupled across the different bidding zones (Single Day Ahead Coupling - SDAC), meaning that the market clearing price is determined simultaneously across all bidding zones[11], taking into account the cross-border transmission capacities and the supply and demand conditions in each zone, which is a crucial aspect for the price formation process and the resulting price dynamics in the Intraday Market, as the latter one is directly linked to the Day-Ahead Market through the trading of the same contracts.

[11] Exact pricing is carried out by the EUPHEMIA algorithm.
[12] Based on [Kie19], as of 27.05.2026, 13:19:50 CET/CEST.

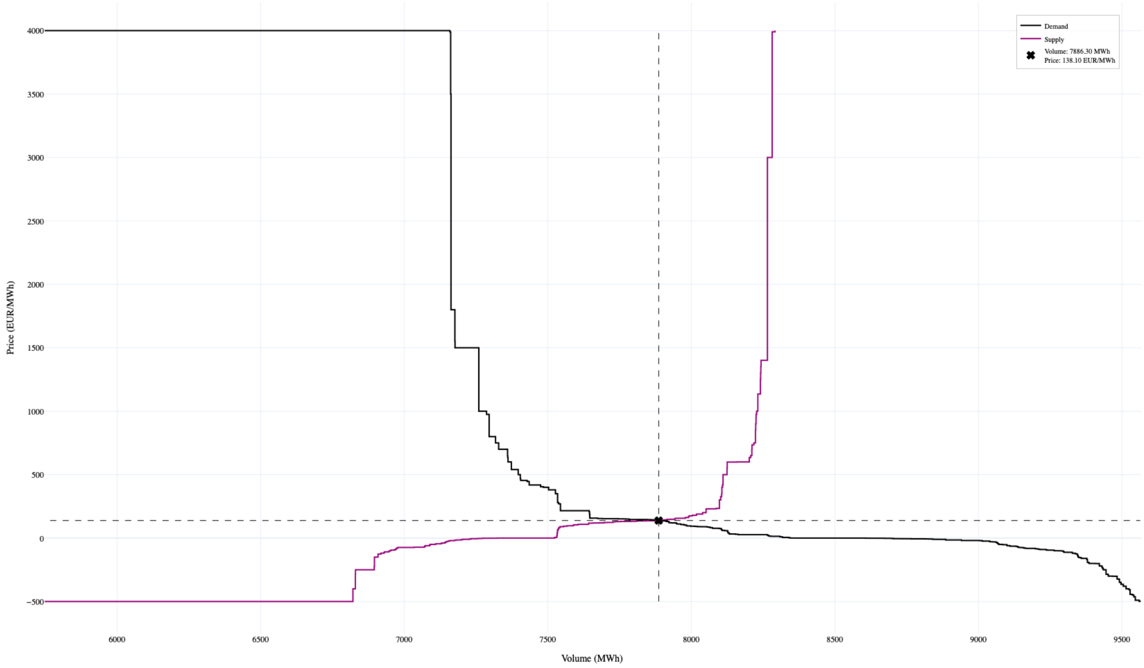


Figure 2.3: Day-Ahead Aggregated Supply and Demand Curves for the Market Area DE-LU, for the Delivery Date: 28th of May, 00:00-00.15.[12]

2.1.3 The Intraday Market

As the physical delivery of the traded contracts is also mandatory on the intraday market, purely financially motivated participants seek to avoid not-netted positions heavily, as they would have to be balanced out in the balancing market, which is the most expensive market, as it is the last market before delivery. The main purpose of the Intraday Market is to balance out the previously locked in Day-Ahead positions that may deviate due to Power Plant Outages, optimisation of power plant usage (generator), optimisation of demand (costumer), renewable energy producer – changes of forecasts[Kie16]. The CID Market starts on the day before delivery (D-1) at 15:00 CET after closing of the Day-Ahead Market with opening the order books for all 24 hours of the next following day resulting in 24 books in parallel (with the hour 01: 00:00 - 01:00 being the first), 30 Minutes and 15 Minutes. The single resolutions can be traded up until 5 Minutes before delivery. Due to this nature the contract lifetime lasts from 8 hours and 55 Minutes (for the first hour of the next following day D) up until 31 hours and 55 Minutes (for the last hour of the next following day D). This natural ordering of the hours over the course of the day, also groups them inherently into “peak’ and “off-peak’ hours, stressing their different dynamics during their contract lifetime for trading. Simplified examples for hourly products can be seen in Figure 2.1, a detailed table listing all hourly specificas can be found in the appendix in Table A.1. The main CID market parameters are summarized below in Table 2.1. The CID market is accompanied by three intraday auctions (IDA), which are single auctions, similar to the Day-Ahead market, that take place at specific times each day:

- **IDA 1**: At 15:00 CET on the day before delivery (D-1).
- **IDA 2**: At 22:00 CET on the day before delivery (D-1).
- **IDA 3**: At 10:00 CET on the day of delivery (D).

These IDA serve the purpose of supporting transparency and liquidity pooling, to provide reference prices along the CID contract lifetimes. Similar as the Day-Ahead market, the CID market is also coupled across the different bidding zones, through the so-called Single Intraday Coupling (SIDC), which is a continuous trading mechanism that is based on the XBID architecture, operating on the M7 system of the EPEX SPOT. The IDA auctions are already part of this mechanism, where the XBID architecture collects all orders in a Shared Order Book (SOB). In a window of 20 Minutes before and after, as well as during the IDA gate-closure the continuous XBID market is shut down. Trading on it for a specific contract is finally shut down 30 Minutes before delivery, still leaving trading open for participants of the same delivery-area up until 5 Minutes before delivery. Its official chronological start is therefore at 15:20 CET at the day before delivery D-1. The XBID related market closing times each day:

- **Around IDA 1**: From 14:40 to 15:20 CET on the day before delivery (D-1).
- **Around IDA 2**: From 21:40 to 22:20 CET on the day before delivery (D-1).
- **Around IDA 3**: From 09:40 to 10:20 CET on the day of delivery (D).

These periods are specifically important, since this harsh reductions in volumes can be also seen in volume plots later on.

Parameter	**Value**
Tick Size	0.01 EUR/MWh
Minimum Price	-9999.9 EUR/MWh
Maximum Price	9999.9 EUR/MWh
Minimum Order Size	0.1 MWh
Maximum Order Size	1000 MWh
Trading Hours	24/7 (with specific gate closure times)
Trading Days	All calendar days (including weekends and holidays)
Settlement Process	Financial settlement based on the market price at delivery time
Price Grid	Discrete price levels determined by the tick size
Time Grid	Hourly, 30-minute, and 15-minute contracts available for trading
Gate Closure Times	Varying based on the delivery hour, with earlier closures for earlier hours (see Figure 2.1)
Order Types Allowed	Limit Orders, Market Orders, and Cancellations

Table 2.1: Key Parameters of the EPEX SPOT CID Market.[13]

[13] Based on: .

2.1.4 The Balancing Market

As bridge to the previous subchapter: Balancing market exists as to balance out the deviations from the locked in Intraday Market volumes

The following table will be provided in the Appendix: Test

2.2 Organisation of Intraday Power Trading

As the main trading takes place on-exchange and is not broker-facilitated, the core price building mechanism is represented by the LOB.[14] Most academic research is centered around data from highly liquid markets, such as equity or forex markets, where the bid-ask spread is usually only the tick-size as the minimum price-increment, that is possible. Therefore these models often only concern the best bid and aks prices, as there is most volume and a huge market-depth, often not entirely depleted by incoming market orders. In stark contrast, the electricity markets are characterized by a much wider bid-ask spread, that can be several times the tick-size, and a much lower market depth, where incoming market orders often deplete the entire volume at the best bid or ask price, leading to a mid-price movement. This unique characteristic of the electricity markets necessitates a different modeling approach that can effectively capture the dynamics of the LOB and the resulting price movements, which is what this thesis focuses on.

2.2.1 Limit Order Books and Trading Mechanics

The LOB is defined on a fixed discrete grid of prices (price levels), where the step size of each price level is called *tick or tick-size*. This tick-size in the CID market is 0.01 EUR/MWh, which means that the price levels are discretized in increments of 0.01 EUR/MWh. The *tick-size* will be denoted by Δp. The respective minimum and maximum volumes for submitted orders are 0.1 MWh and 1000 MWh, which means that market participants can submit orders for quantities within this range. The order-size will be denoted by V.

The trading hours are 24/7, with specific gate closure times that vary based on the delivery hour, as illustrated in Figure 2.1. All together these parameters are called the resolution parameters of a LOB and are summarised in Table 2.1.

Starting from a basic market setup, participants can submit either Limit Orders (LO) or Market Orders (MO). Participants, sending a MO wish to immediately buy or sell their submitted volume at the best price currently available. Due to this unconditional specification, MOs are also referred to as *aggressive orders*, as they are willing to accept the current market price and reduce the resting volumes at the best bid or ask price.

[14] See [Bun25], where only 0.2 TWh of Intraday Trading was traded on broker platforms vs. 102 TWh on EPEX Spot.

Across the multiple, discrete price levels, the best bid price is the highest price at which there is a buy order, while the best ask price is the lowest price at which there is a sell order.

Definition 2.1 (Best bid and ask prices).

$$\text{Best Bid}(t) = \max\{p_i | q_i > 0, p_i < \text{Best Ask}(t)\} \tag{2.1}$$

$$\text{Best Ask}(t) = \min\{p_i | q_i > 0, p_i > \text{Best Bid}(t)\} \tag{2.2}$$

The natural difference between the best bid and ask prices is called the *bid-ask spread*, which represents the cost of trading and is a key indicator of market liquidity. The bid-ask spread is defined as the difference between the best ask price and the best bid price, and it reflects the cost that market participants incur when executing a trade. A narrower bid-ask spread indicates a more liquid market, where there is a smaller cost to trading, while a wider bid-ask spread indicates a less liquid market, where there is a larger cost to trading.

Definition 2.2 ((Quoted) Bid-ask spread).

$$\text{Bid-ask spread}_t = P_t^{\text{bid}} - P_t^{\text{ask}} \tag{2.3}$$

From the full bid-ask spread, the half spread can be derived, which represents the cost of trading for a single side of the market, either buying or selling.

Definition 2.3 ((Quoted) Half spread).

$$\text{Half spread}_t = \frac{1}{2} \cdot \text{Bid-ask spread}_t \tag{2.4}$$

This means that incoming MOs can potentially deplete the entire volume at the best bid or ask price, leading to a subsequent mid-price movement. The mid-price is defined as the arithmetic average of the best bid and ask prices and often treated as a proxy for the true underlying price of the asset.

Definition 2.4 (Mid-price).

$$\text{Mid-price}_t = \frac{1}{2} \cdot (P_t^{\text{bid}} + P_t^{\text{ask}}) \tag{2.5}$$

Definition 2.5 (Market Order (MO)). A *buy* (respective *sell*) market order is

In contrast, participants sending a LO specify a price at which they are willing to buy or sell a certain quantity of electricity, and these orders are added to the LOB and wait for a matching counterparty to execute the trade. LOs are also referred to as *passive orders*, as they provide liquidity to the market by adding volume to the LOB, rather than immediately executing against existing orders.

Definition 2.6 (Limit Order (LO))**.** A *buy* (respective *sell*) limit order is

How the incoming MO are matched with resting LO to result in a trade, is determined by the matching mechanism of the exchange system. Often this algorithm is based on the price-time priority principle, according to which orders are ranked based on their price and the time they were submitted, just as the name already suggests [CJP15]. Then the MO is matched with the best available LO in the LOB, which is the LO with the highest price for a buy MO and the LO with the lowest price for a sell MO. If there are multiple LO at the same price level, the MO is matched with the LO that was submitted first, following the time priority principle. This process continues until the MO is fully executed or there are no more matching LO available in the LOB. When the volume of the MO exceeds the available volume at the best price level, the remaining volume of the MO is then matched with the next best price level, and this process continues until the entire MO is executed or there are no more matching LO available in the LOB. This process is also referred to as *walking the book*.

If a sell-MO is executed against a buy-LO, it is said to *hit the bid*, while if a buy-MO is executed against a sell-LO, it is said to *lift the ask*.

Besides the discussed basic order types, there are also other modalities available:

- **Iceberg Orders**: These are large orders that are split into smaller visible portions, allowing traders to hide the true size of their order from the market.
- **Stop Orders**: These are orders that become active only when a certain price level is reached, allowing traders to limit their losses or lock in profits.
- **Fill or Kill (FOK) Orders**: These are orders that must be executed immediately in their entirety or not at all, ensuring that traders either get the full quantity they want or no trade occurs.
- **Immediate or Cancel (IOC) Orders**: These are orders that must be executed immediately, but any portion of the order that cannot be filled is canceled, allowing traders to get as much of their order filled as possible without waiting for the entire order to be executed.
- **All or None (AON) Orders**: These are orders that must be executed in their entirety, but they do not have to be executed immediately, allowing traders to ensure that they get the full quantity they want without having to worry about partial fills.

- **Market Sweep Orders**: These are orders that are designed to quickly execute against multiple price levels in the LOB, allowing traders to quickly fill large orders without having to worry about the time it takes for the order to be executed.

Here the table of the EU Report with Percentages

The LOB in every state, can essentially be modelled by the two central inputs, that are the arrival of limit orders and market orders, as these trigger the resting LOs and result in trade executions.

Here the mathematical definition of the LOB:

Definition 2.7 (LOB)**.**

$$\mathrm{LOB}(t) = \{(p_i, q_i, t_i) | i = 1, 2, ..., N(t)\} \tag{2.6}$$

$$\mathrm{LOB}(t) = \{(p_i, q_i, t_i) | i = 1, 2, ..., N(t)\} \tag{2.7}$$

where p_i is the price level of the i-th order, q_i is the quantity of the i-th order, t_i is the time of arrival of the i-th order, and $N(t)$ is the total number of orders in the LOB at time t.

Here the derived optimal depths with a surrounding solid line box:

$$\boxed{\begin{aligned} \delta_b^*(q,t) &= \arg\max_{\delta>0} \mathbb{E}\big[U(\mathrm{PNL}_T) \mid q_t = q\big] \\ \delta_a^*(q,t) &= \arg\max_{\delta>0} \mathbb{E}\big[U(\mathrm{PNL}_T) \mid q_t = q\big] \end{aligned}} \tag{2.8}$$

2.2.2 Order types and classification

As already described earlier, orders can be classified into two main categories, market orders and limit orders, based on their execution characteristics. However, within these two broad classes, the ones of special interest for modelling purposes are those that actually move the mid price. To this *aggressive type*, also cancellations belong, as with a cancellation, the respective order loses its position in the price-time priority queue, leaving the next order in line to become the new best bid or ask price.

Here the categorization into 12 types, where we focus on those types tha actually move the mid price, see also the descriptive breakdown or orders of a day and their percentage share.

According to [LV19], the main order types in a limit order book (LOB) can be categorized as can be seen in the table below:

[15] Based on [LV19], as of 16.05.2026.

Order Type	Order Arrival Event	Bid Price Effect	Ask Price Effect
1	aggressive market buy	0	+
2	aggressive market sell	-	0
3	aggressive limit buy	+	0
4	aggressive limit sell	0	-
5	aggressive limit buy cancellation	-	0
6	aggressive limit sell cancellation	0	+
7	non-aggressive market buy	0	0
8	non-aggressive market sell	0	0
9	non-aggressive limit buy	0	0
10	non-aggressive limit sell	0	0
11	non-aggressive limit buy cancellation	0	0
12	non-aggressive limit sell cancellation	0	0

+/0/- indicate whether the order type causes an increase, no change, or decrease in the bid or ask price, respectively.

Table 2.2: Main Order Types in a Limit Order Book.[15]

2.2.3 Matching Mechanism

2.3 The Market Making Problem

Here also the different risk sources, adverse selection, inventory, etc. The classical problem formulation based on Ho & Stoll (1981) is to choose the optimal depth size for posting LO into the book (See [CJP15], page 29)

Also mention the advantage of potentially getting a fee rebate from the exchange in contrast to the taker fee that liquidity takers have to pay for their orders

Key problem: Balancing the optimal limit order posting depth and the fill probabilities

Use stochastic optimal control to tackle the classical problem of solving for the optimal depth as distance from the midprice for posting limit orders, also core problem: how to get the fill probabilities or approximate them

3 Empirical LOB Characteristics and Market Quality

In the following chapter, so called *quality-parameters* of the LOB will be analyzed, which are the key empirical characteristics of the LOB, that will be used to calibrate the mathematical model and to derive the optimal market making strategy. These quality-parameters include

3.1 Data and Data Processing

Here comes a table with the single data columns that can be found:

Here comes a table with the columns: Hour, Year, Number of Days, Average Events per Day:

Hour	Year	Number of (Trading) Days	Average Events per Day
0	2019	365	X
1	2019	365	X
...	...	...	...
23	2019	365	X

Table 3.1: Data Overview: Hourly Event Counts and Averages.

3.2 The Intraday activity pattern: Volumes and Volatilty

Samuleson Effect here From previous intraday pattern: Baseline intensity has to be dependent on the remaining contract lifetime, therefore being non-stationary

Percentage table for the different order types to stress that onyl few types are relevant and also focus on only the best 3 price levels because mostly only 1 tick moved (?).

4 Mathematical Modelling

In order to realistically model the LOB dynamics and derive the optimal market making strategy, we need to choose a suitable mathematical framework that can replicate the key empirical characteristics of the LOB. A natural framework is offered by the Hawkes-Process family, which is a class of self-exciting point processes that can capture the clustering and dependence structure of order arrivals in the LOB. By also including cross-excitation among different order types, this process family also offers modelling capabilities for complex interactions.

In the following sections, we will first introduce the basic concepts of Hawkes processes, and then describe how we can use this framework to model the order arrivals and mid-price dynamics in the electricity markets, ultimately leading to the derivation of the optimal market making strategy.

4.1 The Point Process Family

4.1.1 One Dimensional Hawkes Processes

4.1.2 Multi Dimensional Hawkes Processes

4.2 The Hawkes Process based framework for optimal market making

4.2.1 Excitation Kernels

In the Hawkes-Poission Process, the probability drop-off of an event triggered bya past event is called "kernel".

Here also a plot comparing the different popular choices for the exciattion kernels, but also showing the spline based non-parametric kernel estimation variant.

4.2.2 Time-inhomogeneous (Baseline) Intensities

As we have seen in the empirical analysis, the intraday activity pattern is not constant, but rather follows a certain pattern, with higher activity during certain hours of the day, which is also known as the Samuleson effect. This means that the baseline intensity of the Hawkes process, which represents the underlying rate of order arrivals, has to be dependent on the remaining contract lifetime, therefore being non-stationary. By incorporating time-inhomogeneous baseline intensities into our modeling framework, we can better capture the dynamics of order arrivals and improve the performance of our market making strategy.

4.3 Midprice Dynamics

Here the mathematical definition of the midprice dynamics, which is the central variable of interest in our modeling framework. The

In reference to the before mentioned order type classification, only informed traders post aggressive order types that move the mid price, here from the data again measure the shares of these information carrying order types and show if this problem is immanent and how we can obtain the frequencies from these stats

We need to have the midprice impacted by incoming MO due to realism, cause a certain jump of the process, jump size and direction have to be realistic (in their direction, foresign and magnitude).

4.4 Modelling Order arrivals

Since the LOB and every state can essentially be reconstructed by the two main input parameters, that are the arrival of limit orders and market orders, we can focus on the modelling of these two main order types, as well as cancellations, which are the main drivers of the LOB dynamics. From these general order types, we further focus on those that are actually relevant, i.e. those that move the mid price, which are the so called *aggressive orders*, according to the detailed order-categorization based on XYZ laid out earlier in chapter XYZ. Another central and shaping input is the volume depth of the LOB.[Cho+21]

4.4.1 Market Order Arrivals

4.4.2 Limit Order Arrivals

4.4.3 Cancelations

4.4.4 The role of Order Imbalance

4.4.5 The Complete Model

Although possible in theory, the modelling of cross-excitement among various hourly products is not pursued in this thesis, as the in-parallel lifetime of the contracts differs (See Table) what makes it hard to calculate the cross-excitation, and [Zweiter Grund].

Ignoring the external influence of updated forecasts and other external variables, can be partly defended by seeing that fundamental drivers are partly not relevenat for the liquidity, see here: [HW13].

Also the idea, i.e when we refit daily, we could use these values as prior for the next fit making use of Bayesian updating, see [Jai+24]

4.5 Deriving the optimal strategy: the optimal depths

Here the derivation and solution of the stochastic optimal control problem for the optimal depths, which is

5 Numerical Analysis and Simulation

Since we have to model the midprice dynamics with some form of process, we simulate this process and let the agent follow the optimally derived strategy and then evaluate this simulation study based results

5.1 Exchange Latency

Here the exact calculation of the exchange latency is provided, which is the time it takes for an order to be processed and executed on the exchange after it has been submitted. This includes the time taken for the order to reach the exchange, be processed by the exchange's matching engine, and receive a confirmation of execution. The exchange latency can vary based on several factors, including the type of order, the current market conditions, and the specific exchange being used. Understanding and minimizing exchange latency is crucial for developing effective trading strategies, particularly in high-frequency trading environments where even small delays can impact profitability. important for sampling and fitting durations etc.

5.2 Goodness of Fit Evaluation

QQ Plots here, AIC table, etc.

5.3 Optimal Strategy Backtests

Here some example graph that is shown in the List of Figures:

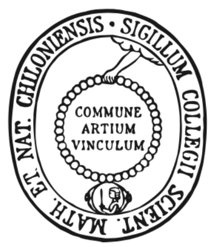


Figure 5.1: Example Figure.[16]

Here some example table that will be shown in the List of Tables section: listing example that will be shown in the List of Algorithms section:

[16] Test cite [LV19], as of 16.05.2026.
[17] See first: [Deu23].

Figure 5.2: Some text.[17]

Column 1	Column 2	Column 3
Data 1	Data 2	Data 3
Data 4	Data 5	Data 6

Table 5.1: Example Table

Listing 5.1: Example Algorithm

```
def example_algorithm(data):
    # Some example algorithm code
    result = data * 2
    return result
```

Her some example algorithm that will appear in the List of Algorithms section:

Algorithm 5.1 Example Algorithm

Require: *data* - Input data for the algorithm
Ensure: *result* - Output result of the algorithm
Initialize *result* to 0
for each element in *data* **do**
    Update *result* based on the element
end for
return *result*

Here some examples of different citation styles offered by LaTeX: Antonio J and Luis A [AL22]. Parenthetical citation: [AL22]. Footnote citation: [AL22]. Footnote citation with page number: [AL22, p. 1]. Inline citation: Rosenberg [Ros03]. Inline citation: Antonio J and Luis A [AL22]. Inline citation: Seabold and Perktold [SP10]. Inline citation: Seabold and Perktold [SP10]. Inline citation: Fama and French [FF16]. Inline citation: Rosenberg [Ros03]. Inline citation: Fama and French [FF16]. Example of a footnote citation.[18].

An example of an equation (numbered and centered) is shown below:

$$E = mc^2 \tag{5.1}$$

[18] [FF16]

6 Literature Review

Here the Literature review text.

6.1 Example Section

7 Methodology

Here the methodology text.

8 Data

Here the data text.

9 Experiments

Here the experiments text.

10 Results

Here the results text.

11 Discussion

Here the discussion text.

12 Conclusion

Here the conclusion text.

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A Supplementary Market Structure Information

A.1 Hourly Products and Trading Durations

Hourly Product	Delivery Interval (CET)	Cross-Border Gate Closure	Max Cross-Border Trading Duration	Local German Gate Closure	Max Local German Trading Duration	Next-Day Hours Open in Parallel	Duration of This Parallel Window
Hour 01	00:00 – 01:00	23:30 (D-1)	8 h 30 min	23:55 (D-1)	8 h 55 min	24	8 h 55 min
Hour 02	01:00 – 02:00	00:30 (D)	9 h 30 min	00:55 (D)	9 h 55 min	23	1 h 00 min
Hour 03	02:00 – 03:00	01:30 (D)	10 h 30 min	01:55 (D)	10 h 55 min	22	1 h 00 min
Hour 04	03:00 – 04:00	02:30 (D)	11 h 30 min	02:55 (D)	11 h 55 min	21	1 h 00 min
Hour 05	04:00 – 05:00	03:30 (D)	12 h 30 min	03:55 (D)	12 h 55 min	20	1 h 00 min
Hour 06	05:00 – 06:00	04:30 (D)	13 h 30 min	04:55 (D)	13 h 55 min	19	1 h 00 min
Hour 07	06:00 – 07:00	05:30 (D)	14 h 30 min	05:55 (D)	14 h 55 min	18	1 h 00 min
Hour 08	07:00 – 08:00	06:30 (D)	15 h 30 min	06:55 (D)	15 h 55 min	17	1 h 00 min
Hour 09	08:00 – 09:00	07:30 (D)	16 h 30 min	07:55 (D)	16 h 55 min	16	1 h 00 min
Hour 10	09:00 – 10:00	08:30 (D)	17 h 30 min	08:55 (D)	17 h 55 min	15	1 h 00 min
Hour 11	10:00 – 11:00	09:30 (D)	18 h 30 min	09:55 (D)	18 h 55 min	14	1 h 00 min
Hour 12	11:00 – 12:00	10:30 (D)	19 h 30 min	10:55 (D)	19 h 55 min	13	1 h 00 min
Hour 13	12:00 – 13:00	11:30 (D)	20 h 30 min	11:55 (D)	20 h 55 min	12	1 h 00 min
Hour 14	13:00 – 14:00	12:30 (D)	21 h 30 min	12:55 (D)	21 h 55 min	11	1 h 00 min
Hour 15	14:00 – 15:00	13:30 (D)	22 h 30 min	13:55 (D)	22 h 55 min	10	1 h 00 min
Hour 16	15:00 – 16:00	14:30 (D)	23 h 30 min	14:55 (D)	23 h 55 min	9	1 h 00 min
Hour 17	16:00 – 17:00	15:30 (D)	24 h 30 min	15:55 (D)	24 h 55 min	8	1 h 00 min
Hour 22	21:00 – 22:00	20:30 (D)	29 h 30 min	20:55 (D)	29 h 55 min	3	1 h 00 min
Hour 23	22:00 – 23:00	21:30 (D)	30 h 30 min	21:55 (D)	30 h 55 min	2	1 h 00 min
Hour 24	23:00 – 24:00	22:30 (D)	31 h 30 min	22:55 (D)	31 h 55 min	1	1 h 00 min

Note: All 24 next-day hours open together at 15:00 CET (D-1). The count decreases as each individual product hits its respective local gate closure.

Table A.1: EPEX Spot CID Trading Durations and Parallel Product Counts.[19]

[19] Based on [Kie19], as of 16.05.2026.

B Data

B.1 LOB Data Description

Column Name	Description
Timestamp	The date and time of the order event
Order Type	The type of order (e.g., limit order, market order, cancellation)
Price	The price at which the order was placed or executed
Quantity	The quantity of the order
Side	The side of the order (buy or sell)
Order ID	A unique identifier for the order
Trader ID	A unique identifier for the trader who placed the order
Venue	The trading venue where the order was placed
Status	The status of the order (e.g., filled, partially filled, canceled)
Time in Force	The duration for which the order is valid (e.g., GTC, IOC)
Execution ID	A unique identifier for the execution event (if applicable)
Counterparty ID	A unique identifier for the counterparty involved in the trade (if applicable)
Order Book State	A snapshot of the limit order book at the time of the event (optional)
Market Conditions	Additional market conditions at the time of the event (optional)
Notes	Any additional notes or comments related to the order event (optional)

Table B.1: Data Columns and Their Descriptions.[20]

Here the first part of the appendix text.

B.2 Second Part

Here the second part of the appendix text.

[20] Based on the typical structure of order book data, as of 16.05.2026.

C Appendix B

D Appendix C

E Code

Here some coding snippets:

Listing E.1: Example Python Code

```python
def example_function(x):
    return x**2
```

F Engineering Configurations

F.1 Hyperparameter Optimization

F.2 Computation Time

Affirmation

I hereby declare that I have composed my Master's thesis *"Optimal Market Making in Intraday Electricity Markets via Hawkes Process Modeling"* independently using only those resources mentioned, and that I have as such identified all passages which I have taken from publications verbatim or in substance. I agree that the work will be reviewed using plagiarism testing software. Neither this paper, nor any extract of it, has been previously submitted to an examining authority, in this or a similar form. I have ensured that the written version of this thesis is identical to the version saved on the enclosed storage medium.

Date, Signature